Wesley W. Erickson

Los Angeles, CA, USA | https://weserickson.com | +1-206-660-5811 | weserickson@gmail.com

PROFILE

Physics PhD with 6 years research experience specialized in stochastic processes, computational physics, and laser-cooled atoms. Experienced with exploratory numerical simulations and developing theoretical models, with strong knowledge of data analysis, statistics, and high performance computing. Multiple years industry experience with software development and data science. Recent familiarity with machine learning techniques and building embedded AI systems. Lab experience includes lasers/optics, electronics design/fabrication, and CAD/machining. U.S. Citizen.

EXPERIENCE

Embedded AI Developer *Yiya Language | Los Angeles, CA | January 2025-Present*

- Led development team for embedded AI-powered language learning device. Developed embedded hardware device, mobile app interface, fine-tuned and optimized LLM and VLM models w/ React, GGML/llama.cpp, PyTorch.
- Presented device and app at US-China international competition in Beijing, project won 1st prize, 100+ finalists.

AI Safety Developer *PIBBSS.ai Fellowship | London, UK | January 2024-September 2024*

- Wrote report showing SGD has heavy tailed gradient noise in both NNs&LLNs, contrary to substantial published work; further demonstrated noise causes "black swan" training events, undermining confidence-interval safety metrics.
- Project involved literature review, exploratory research, designing NN/LLM architectures, building reliable & repeatable training tools, and statistical analysis and visualization.

Research Officer / Full Stack Developer *Vite Labs | Remote | April 2021-September 2022*

- Independently designed and developed three smart contract secured dApps with thousands of unique users and over $1M USD user deposits held.
- Built exchange dashboard with live displays of trading statistics, staking/earning rates, and network health metrics.
- Backend developer for cross chain bridge, developed custodial node contract calls.
- Presented security concerns and proposed solution to existing internal tests to CTO and COO. After approval, built Mocha test suites and tests for internal infrastructure. Tests resulted in discovery of critical security vulnerabilities.
- Primary technical resource, including tech interviews/onboarding (3 hires), 20+ smart contract audits, public tutorial sessions. Taught new hires essential chain features: dApp development, smart contract toolchain & testing framework, high level chain architecture, intro to JS library; focused on security, chain limitations, and differences to Ethereum.
- Maintained multiple backend servers (AWS EC2 dedicated hosts), on-call for outages.

Independent Developer *Independent | Eugene, OR | September 2020-April 2021*

- Built automated node deployment and monitoring tools and interface, operated 28 blockchain nodes for Vite chain.
- Developed and operated arbitrage trading bot for decentralized exchange ~$1000/month profits at peak in Dec 2020.
- Won hackathon bounties from Vite Labs for preliminary smart contract designs, hired full-time in April.

Data Scientist *Covid-19 Polarization Study| Eugene, OR | September 2020-March 2021*

- Analyzed COVID-19 related data including time & content correlations of social media indicators to infection rates.
- Developed new metrics related to echo-chamber effects and topic polarization. Results published in Plos One.

Research Assistant *Atom Optics Lab at University of Oregon | Eugene, OR | September 2014-June 2020*

- Built real-time fullstack interactive experimental console for controlling single-atom experiments with live displays of fluorescence rates, atom counts, parameter settings.

- Developed automated feedback/control systems for single neutral Rb atom release-recapture experiments.
- Designed and implemented low-latency image processing pipeline for EMCCD camera for single atom detection.
- Built FPGA-based photon counter w/ FIFO buffer, embedded Linux, optimized Linux UDP driver.
- Modeled diffusion dynamics of simulated laser-cooled atoms with high performance Fortran and Python/Numba/CUDA based simulations.
- Developed high performance cold-atom diffusion simulations and data processing / exploration tools. Analysis resulted in discovery of novel length scale while investigating conditioned Lévy processes and their relation to extreme events. Results published in Physical Review E.
- Discovered novel length scale while investigating conditioned Lévy processes and their relation to extreme events.
- Built interactive web app for demonstrating Levy statistics and animation demonstrating extreme Levy statistics.
- Explored and modeled diffusion dynamics of laser-cooled atoms through simulations with Fortran, Python, Numba/CUDA.
- Studied conditioned Lévy processes and their relation to extreme events (see [1, 3, 4]).
- Developed feedback/control systems for single neutral Rb atom release-recapture experiments with Perl, C, Verilog (see [3, 5]).

Curriculum Designer / Instructor *S.P.I.C.E. Engineering Camp | Eugene, OR | Summers 2015,2016,2017*

- Developed and taught engineering, electronics, and programming curriculum for 8th grade students.

Senior Reactor Operator / Reactor Operator *Reed Research Reactor | Portland, OR | September 2009-June 2012*

- Licensed by the Nuclear Regulatory Commission to operate and supervise operations at Reed College's TRIGA reactor.
- Responsible for supervising operations, designing & running experiments, training operators, giving oral examinations.
- Prepared and presented lectures on reactor physics.

Full Stack Developer *Code Publishing Company | Seattle, WA | June 2007-August 2008*

- Built text processing pipelines in Perl for parsing municipal code for both print and web publication.
- Built tagging system for indexing and searching large text databases.
- Built secure online portal allowing uploading, updating, and storage for government provided documentation.

EDUCATION

Ph.D. Physics, University of Oregon Eugene, OR – September 2014 to June 2020
Dissertation: Lévy Motion and Laser Cooled Atoms, Advisor: Prof. Daniel A. Steck

M.S. Physics, University of Oregon Eugene, OR – September 2012 to June 2014

B.A. Physics, Reed College Portland, OR – September 2008 to June 2012
Thesis: Electromagnetically Induced Transparency, Advisor: Prof. Lucas Illing

SKILLS

TypeScript, Node.js, Socket.io, Vue.js, Tailwind, Mocha, Solidity, Python, PyTorch, Numba/CUDA, Fortran, C, Golang, Verilog, MongoDB/SQL, Linux, Git, Github Actions, Perl, shell scripts (bash), Docker, AWS, nginx
TypeScript, Node.js, Socket.io, Vue.js, Tailwind, React, Mocha, Solidity, Python, Jupyter, PyTorch, llama.cpp/GGML, train+tune LLMs/ViTs/TTS models, Fortran, C, MongoDB/SQL, Linux, Git/Github Actions, Perl, Docker, AWS, DigitalOcean

Statistical analysis, exploratory analysis, scraping, stochastic processes, random walks, high performance computing, TypeScript, Node.js, Socket.io, Vue.js, Tailwind, D3.js, Matplotlib, Mocha, Python, PyTorch, Jupyter, Numba/CUDA, Fortran, C, MongoDB/SQL, Linux, Git, Github Actions, Perl, shell scripts (bash), Docker

Stochastic processes, Lévy Processes, Itō Calculus, First-hitting-time models, Monte Carlo methods, Statistics, Dynamical Systems, Time Series Analysis, Linux, Git, Docker, KVM Hypervisor, MPI, Profiling, Python, Julia, Fortran, Perl, Mathematica, Matlab/Octave, Numba/CUDA, C (PIC/AVR embedded systems), LabVIEW, KiCad, SolidWorks

ROBOTICS/HARDWARE PROJECTS

- Developed automated feedback/control systems for single neutral Rb atom release-recapture experiments.
- Designed and implemented low-latency image processing pipeline for EMCCD camera for single atom detection.
- Designed and built water-cooled anti-Helmholtz coils designed to trap single neutral atoms (PhD research, University of Oregon)
- Designed and built a single-photon detector, including the lens tube, APD detector, circuitry, and an FPGA with buffering, timestamps, and a TCP stack (PhD research, University of Oregon)
- Developed a wire-grid alpha detector with a USB interface and live Python driven display (graduate E&M project)
- Built and characterized 3D and linear Paul traps, comparing results to COMSOL simulations (undergraduate research at Washington State University)
- Created a LabVIEW-driven, nanometer-precision scanning microscope for measuring quantum dot blinking statistics (undergraduate research at the University of Arkansas)
- Designed and built dot matrix printers from CD drives and hard-drive actuators (as an instructor for the SPICE engineering camp)
- Developed a modular system of detectors and lasers with a CAN bus communication channel, custom C drivers and a USB interface for a “laser maze” activity for children.
- Built a radiation simulator attachment for portable Geiger counters to be used in training (Reed Research Reactor)

TEACHING EXPERIENCE

Graduate Employee *University of Oregon | Eugene, OR | September 2018-June 2020*

- Graded and held office hours for graduate level physics courses.
- AY 2019-2020: Grader for Quantum Mechanics II (PHYS 632, Winter 2020), Quantum Mechanics III (PHYS 633, Spring 2020).
- AY 2018-2019: Grader for Quantum Mechanics I (PHYS 631, Fall 2018), Quantum Mechanics II (PHYS 632, Winter 2019), and Quantum Mechanics III (PHYS 633, Spring 2019).

Graduate Teaching Fellow *University of Oregon | Eugene, OR | September 2018-June 2020*

- Delivered lectures, labs and tutorials for undergraduate level physics courses.
- Participation in course curriculum development (Physics of Sound and Music, Intro to Astronomy)
- Summer 2018: Tutorial Instructor and Grader for General Physics II (PHYS 202) and General Physics III (PHYS 203).
- AY 2017-2018: Research Assistant.
- AY 2016-2017: Tutorial Instructor and Grader for Foundations of Physics I (PHYS 251, Fall 2016), Foundations of Physics II (PHYS 252, Winter 2017), and Foundations of Physics III (PHYS 253, Spring 2017). Lab Instructor for Foundations of Physics Labs (PHYS 290, Fall 2016 and Winter 2017).
- AY 2015-2016: Lab Instructor for Intro Physics Lab I (PHYS 204, Fall 2015), Intro Physics Lab II (PHYS 205, Winter 2016), and Intro Physics Lab III (PHYS 206, Spring 2016).
- Summer 2015: Tutorial Instructor for General Physics II (PHYS 202), and Grader for Physics of Sound and Music (PHYS 152).
- AY 2014-2015: Teaching Assistant for Physics of Sound and Music (PHYS 152, Fall 2014), Grader for Birth and Death of Stars (ASTR 122, Winter 2015), and Grader for Galaxies and the Expanding Universe (ASTR 123, Spring 2015).
- AY 2013-2014: Grader for The Solar System (ASTR 121, Fall 2013), Grader for Birth and Death of Stars (ASTR 122, Winter 2014), and Grader for Galaxies and the Expanding Universe (ASTR 123, Spring 2014).

- AY 2012-2013: Grader for The Solar System (ASTR 121, Fall 2012), Grader for Birth and Death of Stars (ASTR 122, Winter 2013), and Grader for Galaxies and the Expanding Universe (ASTR 123, Spring 2013).
- Teaching Assistant – Reed College Portland, OR – September 2010 to June 2012 Assisted in setup and instruction for 200-level physics labs.

Senior Reactor Operator *Reed Research Reactor | Portland, OR | September 2010-July 2012*

- Licensed by the Nuclear Regulatory Commission. Responsible for supervising operations, designing & running experiments, training operators, giving oral examinations. Prepared and presented lectures on reactor physics.

Senior Reactor Operator *Reed Research Reactor | Portland, OR | September 2009-July 2010*

- Licensed by the Nuclear Regulatory Commission. Responsible running experiments and training operators.

OTHER RESEARCH

Undergraduate Research in Prof. Lucas Illing's lab *Reed College | Portland, OR | September 2011-June 2012*

- Designed and constructed a spectroscopy system used to successfully measure electromagnetically induced transparency in Rubidium atoms (Undergraduate Thesis, see [6]).

Undergraduate Research in Prof. Peter Engels' lab *Washington State University| Pullman, WA | Summer 2011*

- Designed and constructed Paul trap and linear quadrupole trap to trap/image charged lycopodium dust particles.
- Simulated in Mathematica and COMSOL, demonstrated similar dynamics to physical system, extracted parameters.

Undergraduate Research in Prof. Vamsi Komarala's lab *University of Arkansas| Fayetteville, AR | Summer 2010*

- Constructed a confocal laser scanning microscope and successfully measured blinking statistics of individual colloidal quantum dots. Nanometer precision, with a scan routine written in LabVIEW and efficient image analysis in C.

SELECTED PUBLICATIONS

[1] Wesley W. Erickson and Daniel A. Steck, "*Anatomy of an Extreme Event: What Can We Infer About the History of a Heavy-Tailed Random Walk?*", Phys. Rev. E 106, 054142 (2022).
[2] Jun Lang, **Wesley W. Erickson**, Zhuo Jing-Schmidt, "*#MaskOn! #MaskOff! Digital polarization of mask-wearing in the United States during COVID-19*", PLoS One 16(4), e0250817 (2021).
[3] Wesley W. Erickson, "*Lévy Motion and Laser Cooled Atoms*", Ph.D. dissertation, University of Oregon (2020).
[4] Wesley W. Erickson and Daniel A. Steck, "*Lévy Dynamics of Single Laser-Cooled Atoms*", 51st Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, June 1–5, 2020. Portland, OR, USA. Talk abstract: https://meetings.aps.org/Meeting/DAMOP20/Session/J09.3
[5] Richard V. Wagner, **Wesley W. Erickson**, and Daniel A. Steck. "*In situ sensing of position and temperature of a single trapped atom via resonance fluorescence.*" 48th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, June 5–9, 2017. Sacramento, CA, USA. Poster abstract: https://meetings.aps.org/link/BAPS.2017.DAMOP.Q1.68
[6] Wesley W. Erickson, "*Electromagnetically Induced Transparency*", Undergraduate thesis, Reed College (2012).

GRANTS AND AWARDS

- 1st Prize in Main Track at International China US Young Maker Competition, August 2025
- 2024 PIBBSS.ai Fellowship Awardee
- International High Performance Computing Summer School, Toronto, ON, Summer 2015
- NSF Research Experience for Undergraduates, University of Arkansas, Summer 2010